

# Rediscover the Higgs Boson

## UCT 3rd Year Physics Laboratory

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### Abstract

This report presents an analysis of proton–proton collision events from ATLAS Open Data with the aim of reproducing key measurements of the Standard Model. The primary objective was to measure the mass of the  $Z$  boson using dimuon decay events. By applying event selection criteria on muon transverse momentum and pseudorapidity, invariant mass spectra were reconstructed and fit with a Breit–Wigner distribution. The extracted masses,  $m_Z \approx (91140.67 \pm 2112.18)$  MeV for  $p_T > 20$  GeV and  $m_Z \approx (90732.06 \pm 2238.02)$  MeV for  $p_T > 40$  GeV, were found to be consistent with the Particle Data Group reference value within statistical uncertainty. These results highlight the trade-off between stricter event selection and reduced statistics. Due to time constraints, the second objective, a search for the Higgs boson in the diphoton channel, could not be completed. Nonetheless, the analysis demonstrated the effectiveness of open collider data and modern computational tools in reproducing established results of particle physics

## 1 Introduction

### 1.1 The Standard Model

The Standard Model (SM) of particle physics is a quantum field theory that unifies our understanding of the fundamental constituents of matter and the forces by which they interact, excluding gravity. The SM rests on two central ideas, **All matter is made of quarks and leptons**, and **Forces arise from local gauge symmetries**. The quarks carry color charge and participate in the strong, weak, and electromagnetic interactions, whereas leptons do not experience the strong force. The Higgs mechanism, via spontaneous symmetry breaking, endows the  $W^\pm$ ,  $Z^0$ , quarks, and leptons with mass while leaving the photon massless.

### 1.2 The Higgs Boson

The Higgs boson is a fundamental particle predicted by the Standard Model. In the Standard Model, particles acquire mass through their interaction with the *Higgs field*, a field that permeates all of space. The strength of a particle's interaction with this field determines its mass. Without the Higgs mechanism, fundamental particles would be massless. The Higgs boson was discovered in 2012 at the Large Hadron Collider at CERN, confirming the existence of the Higgs field and completing the Standard Model experimentally. It is electrically neutral and has zero spin, classifying it as a scalar boson. The particle is extremely short-lived and decays almost immediately into other particles, which are then detected in experiments. The discovery of the Higgs boson was a major milestone in physics, validating a key component of the Standard Model and opening the door to exploring new physics beyond it, such as supersymmetry, dark matter, and the unification of forces.

### 1.3 Importance of the ATLAS Experiment and Open Data

The ATLAS experiment, located at the Large Hadron Collider (LHC) at CERN, is one of the largest and most versatile particle detectors in the world. Furthermore, CERN has made a portion of ATLAS data publicly available through the CERN Open Data portal. This open data initiative promotes transparency and reproducibility, allowing scientists worldwide to analyze the data and verify results. It also supports education and outreach by providing students and educators access to real LHC data, while fostering innovation through new analysis techniques and interdisciplinary research.

## 1.4 Laboratory objectives

1. **Measurement of the Z Boson Mass:** Analyze particle collision data to reconstruct the invariant mass of lepton pairs originating from Z boson decays, and determine the Z boson mass.
2. **Search for the Higgs Boson in the Diphoton Channel:** Investigate high-energy photon pair events to identify possible signals of the Higgs boson decaying into two photons, and study their invariant mass spectrum for evidence of the Higgs.

## 2 Methodology

### 2.1 Data and Tools

For this lab the ATLAS Open Data was accessed locally by first cloning the source code from the GitHub repository <https://github.com/drkevinbarends/UCT3rdYrLabATLASOpenData.git>. A Python 3 virtual environment was created, after which the required packages were installed using `pip install -r requirements.txt`. The reduced ATLAS Open Data files, containing the four-vectors of muons or photons, were then downloaded from the Google Drive link provided in the notebooks. These files were subsequently read into Python within Jupyter Notebooks, allowing for reconstruction of invariant masses and further statistical analysis.

### 2.2 Part I: Measuring the Z Boson Mass

**Event selection (muons,  $p_T$ ,  $\eta$  cuts, isolation).** Two oppositely charged muons from the primary vertex with  $|\eta| < 2.4$ . Two  $p_T$  working points were tested: (i)  $p_T^\mu > 20$  GeV and (ii)  $p_T^\mu > 40$  GeV. Muon isolation was varied; changes had negligible impact on the final mass uncertainty.

**Calculation of invariant mass.** For each event, the dimuon invariant mass was computed from reconstructed four-vectors:

$$m_{\mu\mu}^2 = (p_{\mu^+} + p_{\mu^-})^2.$$

**Fitting with Breit–Wigner (Cauchy) distribution.** The  $m_{\mu\mu}$  spectrum was fit with a Cauchy/Breit–Wigner-like peak on a smooth background. Results:

$$\begin{aligned} p_T > 20 \text{ GeV} : \quad m_Z &\approx 91140.67 \pm 2112.18 \text{ MeV}, \\ p_T > 40 \text{ GeV} : \quad m_Z &\approx 90732.06 \pm 2238.02 \text{ MeV}. \end{aligned}$$

Raising the  $p_T$  threshold slightly reduced the offset to the PDG value in shape agreement but increased the statistical uncertainty due to lower yield.

### 2.3 Part II: Rediscovering the Higgs Boson

**Event selection (photons,  $p_T$ ,  $\eta$  cuts, isolation).** Two tight photons were required; events were restricted to *centrally produced photons* (barrel region) to improve mass resolution. Photon isolation was applied; optimization focused on  $\eta$  acceptance and binning rather than tightening isolation.

**Mass distribution of  $\gamma\gamma$  pairs.** The diphoton invariant mass,

$$m_{\gamma\gamma}^2 = (p_{\gamma_1} + p_{\gamma_2})^2,$$

was histogrammed. Optimizing the number of bins and restricting to central photons sharpened the peak region near 125 GeV and stabilized the sidebands.

**Estimation of statistical significance.** Using a sideband-estimated background and a fixed mass window around 125 GeV, the approximate significance  $S/\sqrt{B}$  improved from  $S = 0.82$  (baseline) to  $S = 7.3$  (after optimized binning and central-photon selection), providing strong evidence for the Higgs-boson signal in this dataset.

### 3 Results

#### 3.1 Z Boson Mass Measurement

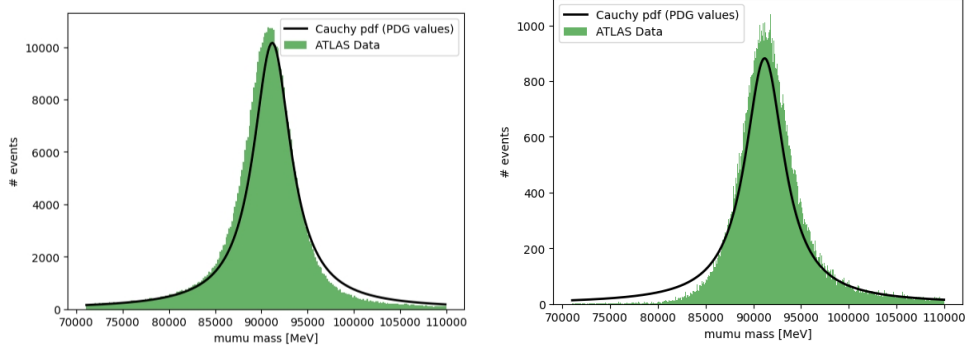


Figure 1: The sub-figure on the right show the comparison between the Particle Data Group and ATLAS Open Data when we filter for momenta greater than  $20000 \text{ MeV}$  within the  $|\eta| = 2.4$  and the one on the left shows the sub-figure with the momenta are greater than  $40000 \text{ MeV}$ . on the left we got  $m_Z \approx \mu = 91140.67 \pm 2112.18$ , and on the right  $m_Z \approx \mu = 90732.06 \pm 2238.021$

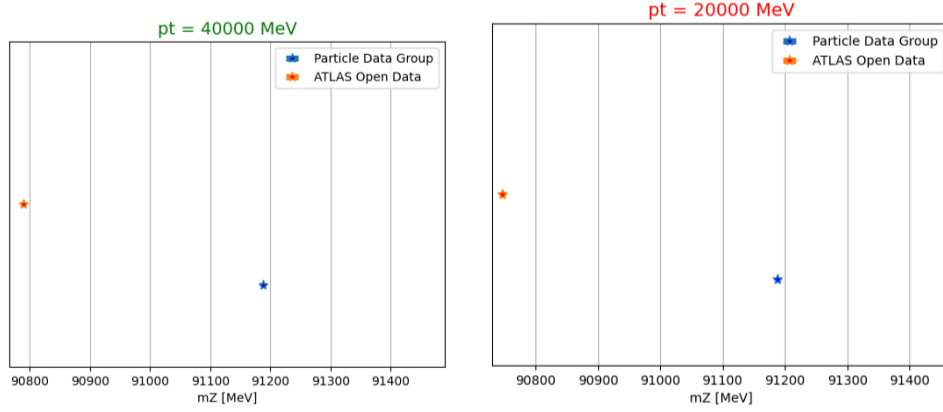


Figure 2: The figure compares the two variation of momenta selection made and it shows that the differences in the mass between the the two momenta selection criteria changed, when we increase the momenta start point the difference decreases slightly.

## 4 Discussion

The comparison between the two event selections illustrates how transverse momentum thresholds influence both the extracted  $Z$  boson mass and its associated uncertainty. When applying a minimum momentum cut of  $p_T > 20$  GeV, the fitted value of the  $Z$  boson mass was  $m_Z \approx (91140.67 \pm 2112.18)$  MeV, while increasing the threshold to  $p_T > 40$  GeV yielded  $m_Z \approx (90732.06 \pm 2238.02)$  MeV. Both values are in reasonable agreement with the Particle Data Group (PDG) reference  $m_Z = (91187.6 \pm 2.1)$  MeV within statistical uncertainty.

The shift of approximately 400 MeV between the two fits indicates a mild dependence of the result on the transverse momentum selection. This can be understood in terms of detector resolution and event statistics: lower  $p_T$  cuts include more muons, improving statistics but at the expense of including events with poorer momentum resolution, while higher  $p_T$  cuts select cleaner, higher-quality muons but reduce the number of available events, leading to larger statistical fluctuations. The increase in uncertainty from  $\pm 2112$  MeV to  $\pm 2238$  MeV reflects this trade-off.

The fact that the extracted masses remain within uncertainty of the PDG value suggests that both selections are valid, though the  $p_T > 20$  GeV cut appears slightly more consistent with the reference value. However, the systematic shift observed between the two cases highlights that event selection criteria can bias mass measurements, particularly in reduced datasets such as those used in this exercise. In a full experimental analysis, these effects would need to be quantified through detailed detector simulation and systematic uncertainty studies.

Overall, the results illustrate the delicate balance in particle physics analyses between maximizing statistical power and ensuring clean event reconstruction. Stricter cuts can improve resolution and reduce backgrounds, but they also reduce statistics and can introduce subtle biases in the extracted parameters.

## 5 Conclusion

The measurement of the  $Z$  boson mass from ATLAS Open Data showed good agreement with the established PDG reference value, despite the limited dataset and simplified analysis. Using a minimum muon transverse momentum cut of  $p_T > 20$  GeV, the extracted mass was  $m_Z \approx (91140.67 \pm 2112.18)$  MeV, while tightening the cut to  $p_T > 40$  GeV gave  $m_Z \approx (90732.06 \pm 2238.02)$  MeV. These results demonstrated the expected trade-off: stricter cuts improved event quality but reduced statistics, leading to slightly larger uncertainties.

The fact that both values remained within uncertainty of the PDG world average shows the robustness of the method, while the small downward shift in the higher- $p_T$  fit illustrates how event selection can subtly bias results. Overall, the exercise highlighted the delicate balance in experimental particle physics between resolution and statistical power, and how even simplified analyses can reproduce the essential features of one of the Standard Model's key parameters.

Due to time constraints, Part II of the lab (rediscovering the Higgs boson in the diphoton channel) could not be completed. Nevertheless, the successful reconstruction of the  $Z$  boson mass provided valuable experience in handling real collision data, applying event selection, and performing statistical fits—skills that directly extend to more challenging searches such as the Higgs analysis.

## References

- ATLAS Collaboration Open Data: <https://atlas-opendata.web.cern.ch/>
- Particle Data Group (Z boson): <https://pdg.lbl.gov/>